

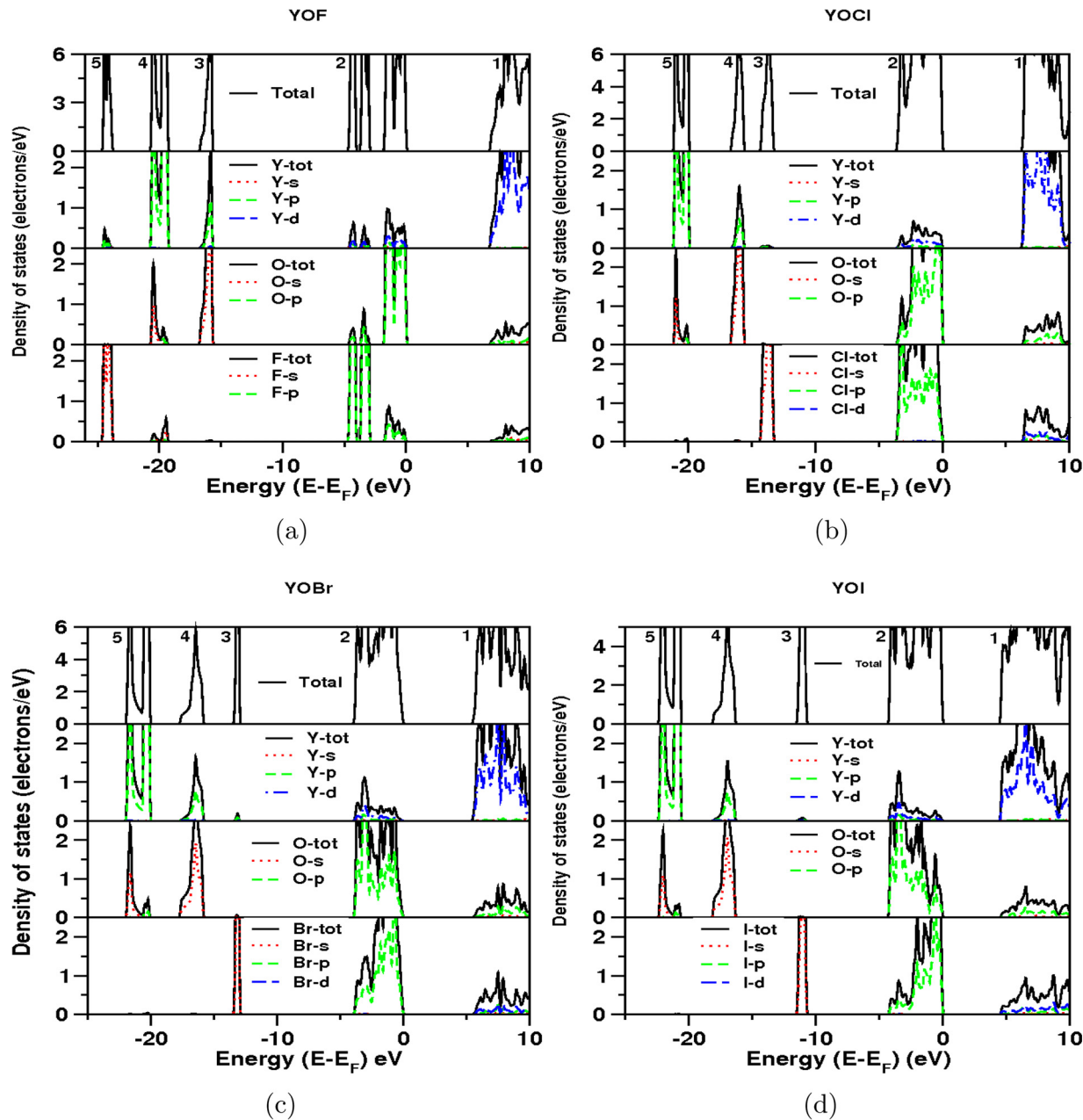


FIG. 3. Partial density of states of YOX (X = F, Cl, Br, and I) compounds calculated using TB-mBJ functional with WIEN2k code. The valence bands are mainly due to hybridized O-*p* and F-*p* states whereas the conduction band is due to Y-*d* states in all the compounds.

red shift of the phonon peak, particularly in the low energy region. The frequencies become smaller on moving from YOF to YOI which is due to increase in the mass of the halides.

All the computed mode frequencies were real. Absence of modes with imaginary frequency clearly shows that all the

compounds studied here are dynamically stable. $E_g^{(1)}$, $E_g^{(2)}$, and $E_g^{(3)}$ modes correspond to motion of Y, X (X = F, Cl), and O atoms, respectively, in case of YOF, YOCl whereas in case of YOBr, YOI these modes correspond to displacement of the X (X = Br, I), Y, and O atoms. Similarly the vibrational modes $A_{1g}^{(1)}$, $A_{1g}^{(2)}$ and $A_{1g}^{(3)}$ belong to the antisymmetric vibration of Y, F, and O atoms, respectively, in case of YOF, whereas in case of YOX (Cl, Br, I) $A_{1g}^{(1)}$ and $A_{1g}^{(2)}$ modes correspond to antisymmetric vibration of Y and X (Cl, Br, and I). The B_{2g} mode corresponds to antisymmetric vibration of O atoms.

D. Optical properties

The knowledge of both the real and the imaginary parts, respectively ϵ_1 and ϵ_2 , of the dielectric function allows for

TABLE II. Elastic constants (C_{ij} , in GPa) of YOX (X = F, Cl, Br, and I) calculated at the theoretical equilibrium volume within LDA.

Compound	C_{11}	C_{12}	C_{13}	C_{14}	C_{33}	C_{44}	C_{66}
YOF	258.5	108.8	105.7	23.8	240.5	94.6	...
YOCl	178.4	59.1	70.3	...	114.4	37.0	72.8
YOBr	157.5	40.4	21.5	...	21.6	20.9	57.7
YOI	143.7	29.6	16.6	...	20.2	8.5	48.4